

Doubly Robust Estimation of Group Effect under Geographic Heterogeneity

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Motivation

- Group disparities are systematic differences in outcomes across population subgroups, a central topic in social science, economics, and public health.

- Group disparity decomposition:

$$D^{(O)}(Y) = \underbrace{\{E_{G_1}(Y) - E_{G_1}(Y | G_2)\}}_{D^{(U)}(Y)} + \underbrace{\{E_{G_1}(Y | G_2) - E_{G_2}(Y)\}}_{D^{(E)}(Y)}$$

$$E_{G_1}(Y | G_2) = \int E(Y | \mathbf{x}, \mathbf{s}, G_2) g(\mathbf{x}, \mathbf{s} | G_1) d(\mathbf{x}, \mathbf{s})$$

“What would happen to individuals in G_1 if they were treated like individuals in G_2 with the same covariates and locations?”

- Y : Outcome (e.g., disease incidence; wages)
- G_1 : Group of interest; G_2 : Reference group; $\mathbf{x}$: Individual-level Covariates; $\mathbf{s}$: Locations
- Two existing approaches:
 - Peters–Belson (PB)**: models the outcome, $E(Y | \mathbf{x}, \mathbf{s}, G_2)$
 - Inverse-probability weighting (IPW)**: models group membership, $P(G_1 | \mathbf{x}, \mathbf{s})$
- Either model yields approximately unbiased estimates only if it is correctly specified

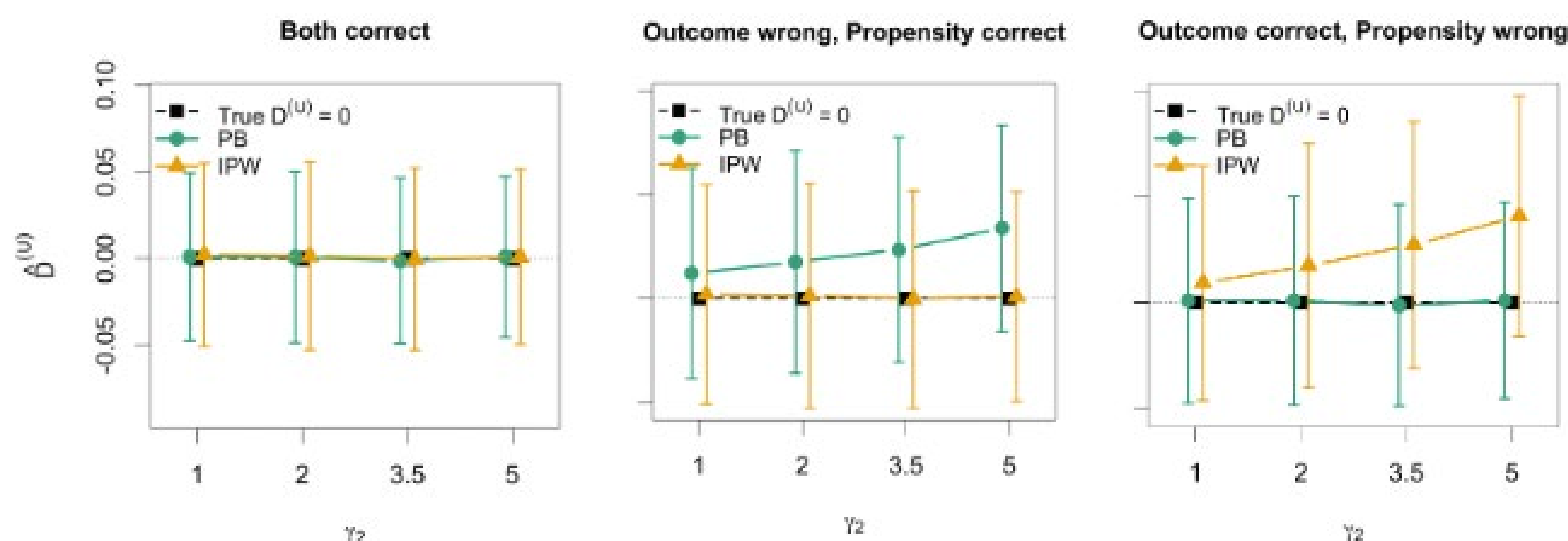


Figure 1. γ_2 controls how different the covariate distribution of x_2 is between two subpopulations. Any single model is sensitive to model mis-specification.

Objectives

The goals of this study are to:

- Develop a doubly robust estimator of group effect that is consistent if either the outcome or the propensity model is correctly specified.
- Capture geographic heterogeneity via bivariate spline smoothing over triangulation (BPST), accommodating sparse and irregularly distributed locations.

Doubly Robust Group-Effect Estimator

Two working geo-additive models:

- Outcome model (Fitted on G_2 in the sample S):

$$\text{logit}(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta} + \mathbf{B}(\mathbf{s}_i)^\top \boldsymbol{\theta}, \quad i \in G_2 \cap S$$

$\mathbf{B}(\mathbf{s})$: bivariate Bernstein basis over triangulation Δ

- Propensity model (Fitted on the sample S):

$$\text{logit}(\pi_i) = \mathbf{x}_i^\top \boldsymbol{\gamma} + \mathbf{B}(\mathbf{s}_i)^\top \boldsymbol{\vartheta}, \quad i \in S$$

Doubly Robust group-effect estimator:

$$\hat{D}_{DR}^{(U)} = \underbrace{\frac{\sum_{i \in S} w_i R_{1,i} (y_i - \hat{\mu}_{G_2,i})}{\sum_{i \in S} w_i R_{1,i}}}_{\hat{D}_{PB}^{(U)}} - \underbrace{\frac{\sum_{i \in S} w_i R_{2,i} \hat{w}_i (y_i - \hat{\mu}_{G_2,i})}{\sum_{i \in S} w_i R_{2,i} \hat{w}_i}}_{\hat{\Delta}_{IPW}^{(U)}}$$

- $\hat{D}_{PB}^{(U)}$: Peters-Belson group effect estimator; $\hat{\Delta}_{IPW}^{(U)}$: propensity-weighted augmentation term
- w_i : sampling weight for $i \in S$; $\hat{w}_i$: propensity odds weight

Theorem: Under regularity conditions, $\hat{D}_{DR}^{(U)}$ is consistent if either $\mu(\mathbf{x}, \mathbf{s})$ or $\pi(\mathbf{x}, \mathbf{s})$ is correctly specified.

Preliminary Data Analysis: Mortgage Denial Rate

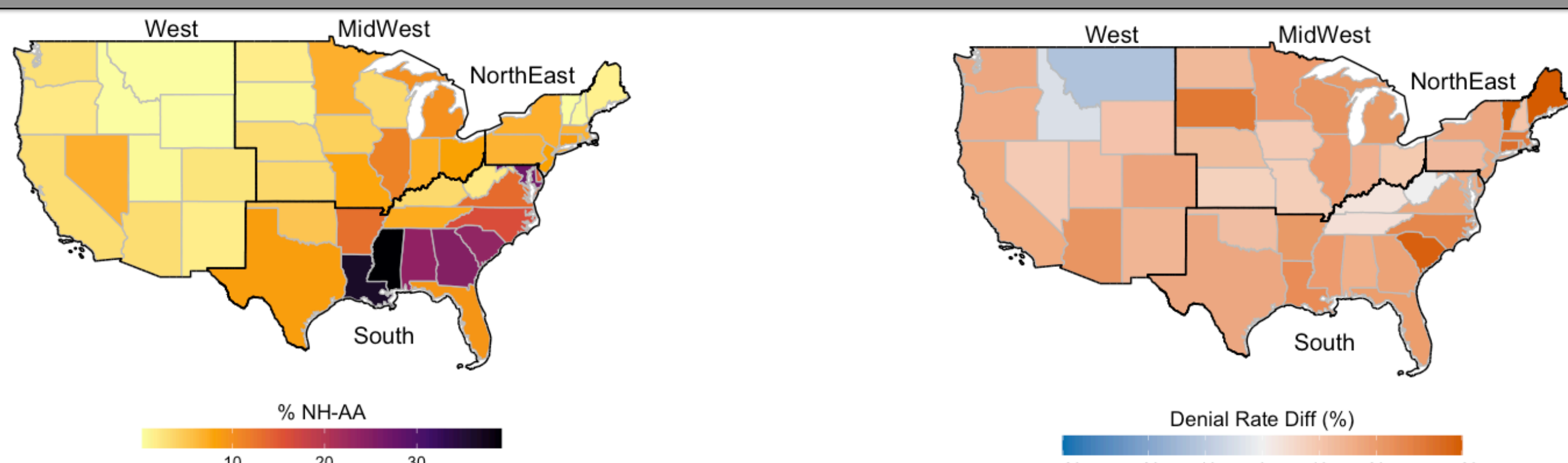


Figure 2. Geographic heterogeneity in 2024 HMDA mortgage denial rate: (left) share of NH-AA applicants by state; (right) raw NH-AA vs. NH-White denial-rate difference (%). The disparity varies markedly across states and is concentrated in the South. Detailed definitions are in the following simulation section.

Data-Driven Simulation

Variables:

- Data: 2024 Home Mortgage Disclosure Act (HMDA) Data
- Outcome of interest Y : denial of a conventional mortgage application (1 = denied, 0 = approved)
- Group membership: G_1 = NH-AA, G_2 = NH-White (contiguous U.S.)
- Covariates $\mathbf{x}$: income, debt-to-income (DTI), area median income
- Locations $\mathbf{s}$: latitude, longitude of individuals
- True coefficients $\boldsymbol{\beta}_0, \boldsymbol{\gamma}_0$ and spatial surfaces s_y, s_g : estimated from HMDA Data
- Generate outcome Y and group membership, $R_{1,i}=1$ represents G_1 , $P(R_1 = 1) \approx 0.35$:

$$Y_i \sim \text{Bernoulli}(p_i), \quad \text{logit}(p_i) = \mathbf{x}_i^\top \boldsymbol{\beta}_0 + s_y(\mathbf{s}_i)$$

$$R_{1,i} \sim \text{Bernoulli}(\pi_i), \quad \text{logit}(\pi_i) = \mathbf{x}_i^\top \boldsymbol{\gamma}_0 + s_g(\mathbf{s}_i)$$

- Identical outcome model across groups, with group membership depending on $(\mathbf{x}, s_g)$ via the propensity model $\Rightarrow D_{true}^{(U)} = 0, D_{true}^{(E)} \neq 0$

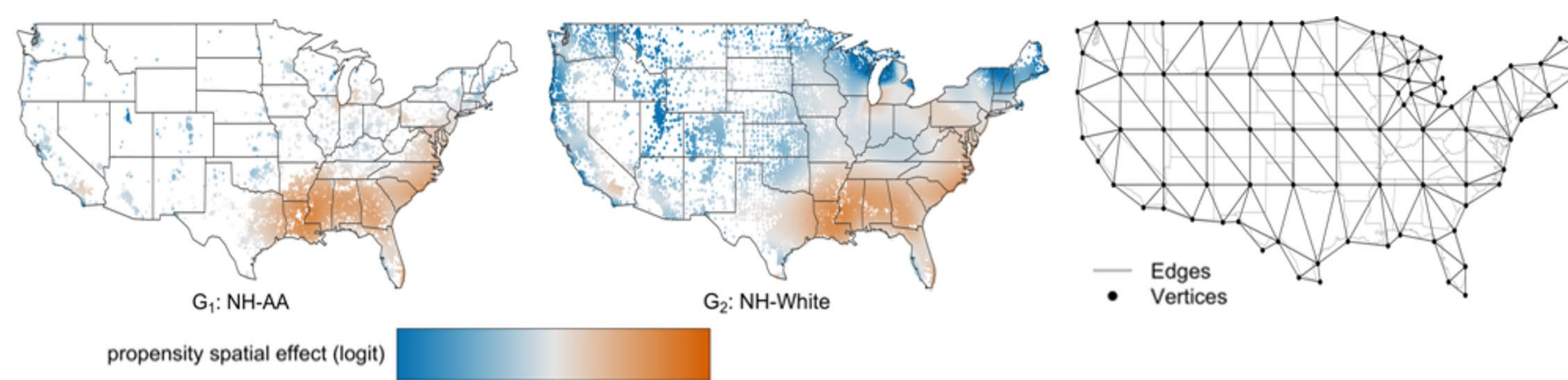


Figure 3. Geographic heterogeneity in group membership: the estimated propensity spatial effect across G_1 and G_2 residence locations (finite population), with the bivariate-spline triangulation used for smoothing.

Controlled model specification:

- A quadratic term of DTI (x_2^2) is injected in the true outcome and propensity models:
 - each working model is correctly specified when it includes x_2^2 and mis-specified when omits x_2^2 , giving four different scenarios
- γ_2 tunes the strength of x_2^2 in the true propensity model, controlling degree of model mis-specification (larger γ_2 = stronger model mis-specification)
- PB, IPW, and DR group-effect estimators compared over 500 replications ($n = 2,000$, SRS) with 95% CIs (Taylor-linearization variance), evaluated on bias and efficiency

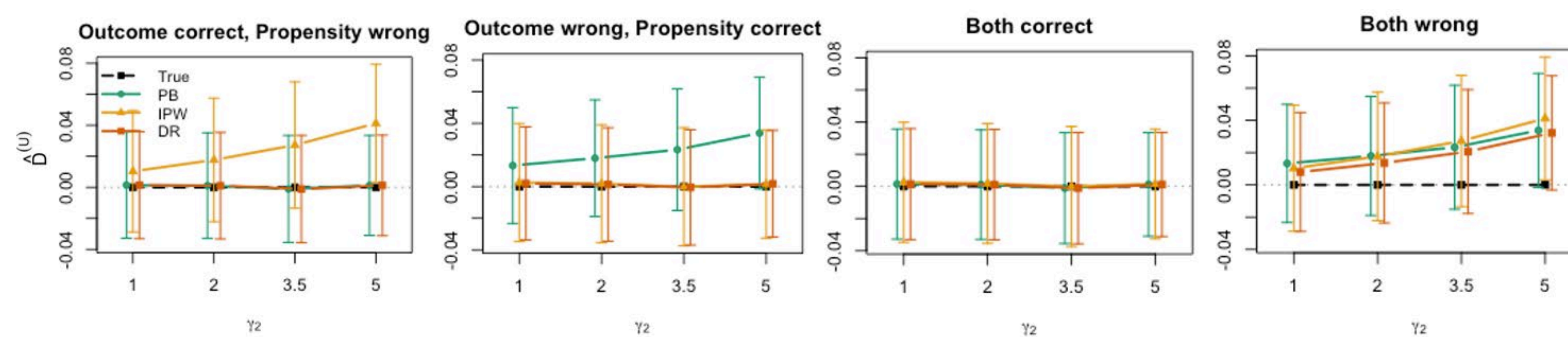


Figure 4. DR remains unbiased whenever either model is correctly specified: estimated $\hat{D}^{(U)}$ vs. γ_2 under four model specification scenarios. DR is also more efficient than IPW where both are unbiased.

- For each state v , we test $H_0: D_v^{(U)} = 0$ with a Wald test ($\alpha = 0.05$) using the DR group-effect estimator and its Taylor-linearization variance:
 - Across all four scenarios no state is significant, confirming $D_v^{(U)} = 0$, which is also consistent with the simulation set-up: $D_{true}^{(U)} = 0$

Taylor-linearization variance:

$$\widehat{\text{Var}}(\hat{D}_v^{(U)}) = \frac{n}{n-1} \sum_{i \in S} \left(\hat{\phi}_{v,i} - \bar{\phi}_v \right)^2, \quad \hat{\phi}_{v,i} \text{ is the influence function of unit } i \text{ in state } v$$

Real Data Analysis

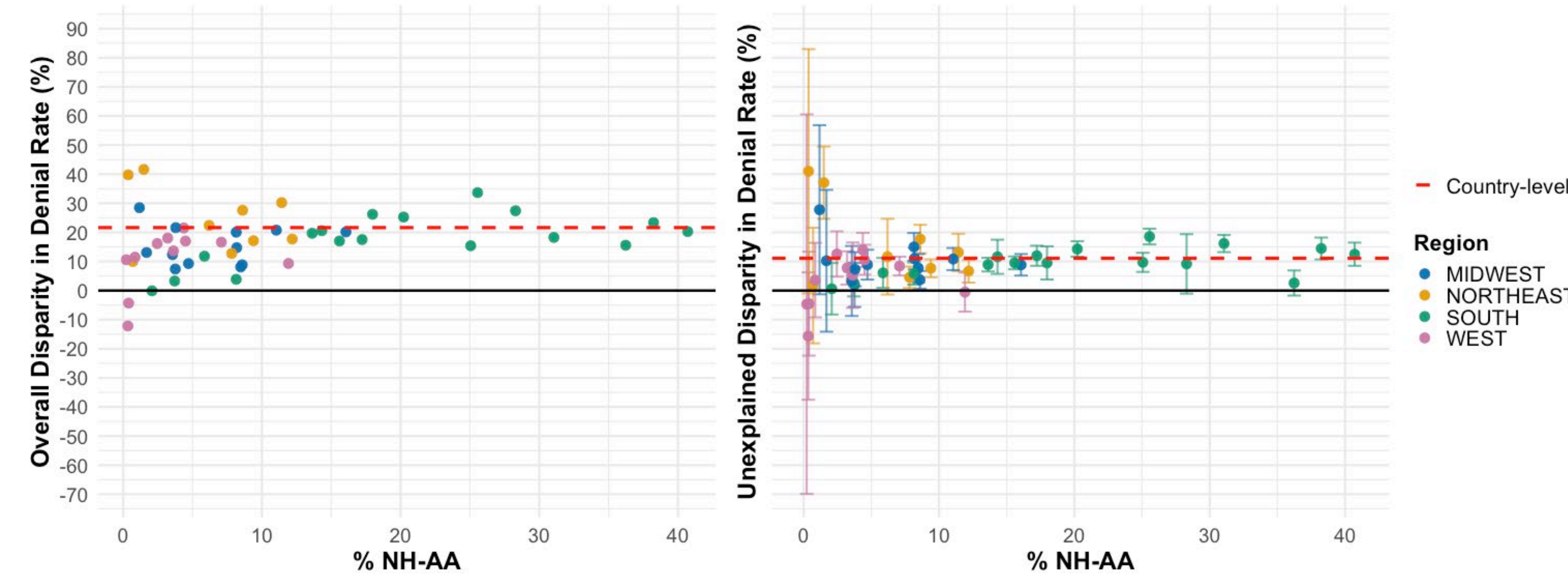


Figure 5. State-level NH-AA vs. NH-White mortgage-denial disparity: overall disparity (left, national ≈ 0.216) and $\hat{D}_{DR}^{(U)}$ (right, national ≈ 0.111) with 95% CIs, against each state's share of NH-AA applicants. Roughly half the overall national gap can be explained after adjusting for covariates and spatial effect. $\hat{D}_{DR}^{(U)}$ also clusters more tightly around the national line than the overall gap (precision-weighted cross-state SD reduced from 0.069 to 0.047), indicating reduced geographic heterogeneity in the group effect.

Covariates:

- Demographic: applicant age, applicant sex
- Socioeconomic: income, debt-to-income (DTI), loan term, lien status
- Neighborhood/area: tract minority share, area median income
- Locations:** individual latitude and longitude